Exact tensor-product boundariness for a two-parameter family of non-unital qubit channels

Xiaojian Zeng
PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

The boundariness of an interior quantum channel is the reciprocal of a maximally entangled numerical radius of its inverse Choi state. Its multiplicativity under tensor products was conjectured from numerical evidence. We prove exact multiplicativity for a two-parameter family of full-rank qubit channels on $|s|, |t| \leq 1/5$, including a relative open subset in which both channels are non-unital. Parameter-dependent positive-semidefinite multipliers bound every entangling $U(4)$ comparator and are saturated by the identity. Six rational LDL^{T} pivots certify positivity on the full intervals. The joint optimization consequently reduces to two closed formulas. The universal arbitrary-channel conjecture remains unresolved.

Keywords: quantum channels; Choi states; boundariness; non-unital noise; tensor-product multiplicativity

1 Introduction

Completely positive trace-preserving maps form a compact convex set once the input and output dimensions are fixed. The *boundariness* of an interior channel measures the smallest channel weight needed in a convex decomposition that reaches the boundary. Through the Choi–Jamiołkowski representation [1, 2], this is a question about positive matrices with a fixed marginal. It also has an operational interpretation through minimum-error discrimination: the channel farthest from a given interior channel can be chosen unitary [3].

Puchała, Jenčová, Sedlák, and Ziman derived an inverse-Choi formula for channel boundariness, proved submultiplicativity under parallel tensor products, and proved equality when one factor is the completely mixed replacement channel [3]. They reported numerical evidence for general multiplicativity but left it as a conjecture. The issue is whether an entangling unitary comparator for a product channel can outperform all product comparators.

This article gives an exact positive result on a deliberately restricted domain. We construct two one-parameter families of full-rank qubit channels and prove multiplicativity on their complete parameter square. Away from the coordinate axes both channels are non-unital, so the result is not the known completely mixed replacement case. The proof uses tight one-sided positive-semidefinite multipliers for the two local inverse Choi matrices. These local certificates tensor to control every unitary in $U(4)$. Their positivity is established by rational interval inequalities, without a grid, floating-point optimizer, or perturbative truncation.

The result does not prove or disprove multiplicativity for arbitrary full-rank qubit channels. Its value is instead twofold: it supplies a continuum with two genuinely non-unital factors, and it isolates a reusable certificate that can replace joint-unitary optimization when suitable local multipliers exist.

2 Normalized Choi convention and the tensor question

Let $A \simeq B \simeq \mathbb{C}^d$ denote the input/reference and output spaces. We use the normalized maximally entangled vector and normalized Choi state

$$|\Omega_d\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j\rangle_B |j\rangle_A, \qquad J_{\mathcal{G}} = (\mathcal{G} \otimes \mathrm{id}_A)(|\Omega_d\rangle\langle\Omega_d|). \tag{1}$$

Thus $J_{\mathcal{G}} \succeq 0$, $\operatorname{Tr} J_{\mathcal{G}} = 1$, and $\operatorname{Tr}_B J_{\mathcal{G}} = I_A/d$. The basis order is always output followed by input, (B, A).

For a $d \times d$ matrix V, define row vectorization by

$$\operatorname{rvec}(V) = \sum_{b,a} V_{ba} |b\rangle_B |a\rangle_A, \qquad |u_V^{(d)}\rangle = \frac{\operatorname{rvec}(V)}{\sqrt{d}}. \tag{2}$$

For Hermitian A on $B \otimes A$, write

$$R_d(A) = \max_{U \in U(d)} \langle u_U^{(d)} | A | u_U^{(d)} \rangle. \tag{3}$$

If $J_{\mathcal{G}} > 0$, the formula of Ref. [3], converted from its unnormalized vector $|U\rangle\!\rangle = \sqrt{d}\, |u_U^{(d)}\rangle$, is

$$b(\mathcal{G}) = R_d(J_{\mathcal{G}}^{-1})^{-1}. \tag{4}$$

Equivalently, this is the quantity defined directly by

$$b(\mathcal{G}) = \inf_{\mathcal{C}\ \mathrm{CPTP}} \max\{q \in [0,1] : J_{\mathcal{G}} - q J_{\mathcal{C}} \succeq 0\}. \tag{5}$$

For two qubit channels, the raw tensor order is $B_1 A_1 B_2 A_2$. Let P be the permutation unitary defined by

$$P|b_1, a_1, b_2, a_2\rangle = |b_1, b_2, a_1, a_2\rangle. \tag{6}$$

Then the normalized Choi state of one parallel use is

$$J_{\mathcal{E}\otimes\mathcal{F}} = P(J_{\mathcal{E}} \otimes J_{\mathcal{F}})P^\dagger. \tag{7}$$

Consequently, with $A = J_{\mathcal{E}}^{-1}$ and $B = J_{\mathcal{F}}^{-1}$, multiplicativity of boundariness is equivalent to

$$R_4\Big(P(A \otimes B)P^\dagger\Big) = R_2(A) R_2(B). \tag{8}$$

Product unitaries give the "$\geq$" direction in (8), or equivalently the established submultiplicativity $b(\mathcal{E} \otimes \mathcal{F}) \leq b(\mathcal{E}) b(\mathcal{F})$. The reverse inequality must control all entangling $V \in U(4)$.

For a qubit, the magic-basis reduction used in Ref. [3] gives

$$R_2(A) = \lambda_{\max}\Big(\operatorname{Re}(W^\dagger A W)\Big), \qquad W = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 & i \\ -1 & i & 0 & 0 \\ 1 & i & 0 & 0 \\ 0 & 0 & 1 & -i \end{pmatrix}, \tag{9}$$

where Re is entrywise real part. This realizes the maximally entangled numerical range as a real numerical range; see also Ref. [4].

3 A tensorizable multiplier criterion

The following elementary criterion is the mechanism behind the family result.

Proposition 1 (Tight local multipliers)**.** *Let $A, B > 0$ act on two copies of $\mathbb{C}^2 \otimes \mathbb{C}^2$. Suppose there are Hermitian 2×2 matrices L_A, L_B such that*

$$\frac{A}{2} \preceq I_2 \otimes L_A^{\mathsf{T}}, \qquad \frac{B}{2} \preceq I_2 \otimes L_B^{\mathsf{T}}, \tag{10}$$

and

$$\operatorname{Tr} L_A = R_2(A), \qquad \operatorname{Tr} L_B = R_2(B). \tag{11}$$

Then (8) holds.

Proof. For row vectorization,

$$\operatorname{rvec}(V)^\dagger (I_d \otimes L^{\mathsf{T}}) \operatorname{rvec}(V) = \operatorname{Tr}(LV^\dagger V). \tag{12}$$

Tensoring the two positive inequalities in (10) and applying P gives

$$\frac{P(A \otimes B)P^\dagger}{4} \preceq I_4 \otimes (L_A \otimes L_B)^{\mathsf{T}}. \tag{13}$$

Indeed, if $Q_A = I_2 \otimes L_A^{\mathsf{T}} - A/2$ and $Q_B = I_2 \otimes L_B^{\mathsf{T}} - B/2$, the difference on the right of (13) is the permutation of

$$Q_A \otimes (I_2 \otimes L_B^{\mathsf{T}}) + (A/2) \otimes Q_B, \tag{14}$$

which is positive semidefinite. Hence every $V \in U(4)$ obeys

$$\langle u_V^{(4)} | P(A \otimes B)P^\dagger | u_V^{(4)} \rangle \le \operatorname{Tr}\Big[(L_A \otimes L_B)V^\dagger V\Big] \tag{15}$$

$$= \operatorname{Tr}(L_A) \operatorname{Tr}(L_B) = R_2(A)R_2(B). \tag{16}$$

The reverse bound is attained by a tensor product of local maximizing unitaries. This proves the claim. □

The criterion is a sufficient condition, not an assertion that tight multipliers exist for every qubit channel. In the family below they can be written in closed form and certified exactly.

4 The non-unital two-parameter family

Let $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $Z = \operatorname{diag}(1, -1)$, and use the matrix W in (9). Define

$$J_E(s) = W \operatorname{diag}\left(\frac{1}{4}, \frac{3}{10}, \frac{1}{10}, \frac{7}{20}\right) W^\dagger + \frac{s}{4}(Z \otimes I_2), \tag{17}$$

$$J_F(t) = W \operatorname{diag}\left(\frac{3}{10}, \frac{1}{5}, \frac{1}{10}, \frac{2}{5}\right) W^\dagger + \frac{t}{4}(X \otimes I_2). \tag{18}$$

Lemma 2 (Physical admissibility)**.** *For every real $|s|, |t| \le 1/5$, $J_E(s)$ and $J_F(t)$ are normalized, positive-definite, trace-preserving qubit Choi states. More precisely,*

$$\lambda_{\min}(J_E(s)), \lambda_{\min}(J_F(t)) \ge \frac{1}{20}. \tag{19}$$

Their output marginals are

$$\operatorname{Tr}_A J_E(s) = \frac{I_2 + sZ}{2}, \qquad \operatorname{Tr}_A J_F(t) = \frac{I_2 + tX}{2}. \tag{20}$$

Thus the corresponding channels are both non-unital whenever $st \neq 0$.

Proof. The four unperturbed eigenvalues in each definition are positive, sum to one, and have minimum $1/10$. The perturbations have trace zero and vanish under Tr_B, so $\mathrm{Tr}_B J_E = \mathrm{Tr}_B J_F = I_2/2$. Their operator norms are $|s|/4$ and $|t|/4$, at most $1/20$. Weyl's inequality gives (19). Taking Tr_A gives (20); in the normalized convention this marginal is the channel output on $I_2/2$. □

Exact inversion and the qubit formula (9) give

$$\mathrm{Re}(W^\dagger J_E(s)^{-1} W) = \mathrm{diag}\left(\frac{24}{6-5s^2}, \frac{20}{6-5s^2}, \frac{140}{14-25s^2}, \frac{40}{14-25s^2}\right), \tag{21}$$

$$\mathrm{Re}(W^\dagger J_F(t)^{-1} W) = \mathrm{diag}\left(\frac{160}{48-25t^2}, \frac{40}{8-25t^2}, \frac{80}{8-25t^2}, \frac{120}{48-25t^2}\right). \tag{22}$$

On the complete parameter intervals the third entry is strictly largest in both lines. The third column of W is $\mathrm{rvec}(I_2)/\sqrt{2}$, so the local maximizer is the identity up to phase and

$$R_E(s) = \frac{140}{14-25s^2}, \qquad R_F(t) = \frac{80}{8-25t^2}. \tag{23}$$

Theorem 3 (Exact multiplicativity on the closed square)**.** *Let $\mathcal{E}_s$ and $\mathcal{F}_t$ be the qubit channels with normalized Choi states (17) and (18). For all real $|s|, |t| \le 1/5$,*

$$b(\mathcal{E}_s \otimes \mathcal{F}_t) = b(\mathcal{E}_s) b(\mathcal{F}_t) = \frac{(14-25s^2)(8-25t^2)}{11200}. \tag{24}$$

Equivalently, every $V \in U(4)$ is bounded in (8) by $R_E(s)R_F(t)$, and $V = I_4$ attains the bound. The saturating joint comparator is unique up to an overall phase.

Proof. Set $A = J_E(s)^{-1}$ and $B = J_F(t)^{-1}$. Define

$$L_E(s) = \frac{10}{14-25s^2}\begin{pmatrix} 7-5s & 0 \\ 0 & 7+5s \end{pmatrix}, \qquad L_F(t) = \frac{1}{8-25t^2}\begin{pmatrix} 40 & -50t \\ -50t & 40 \end{pmatrix}. \tag{25}$$

Both matrices are positive definite on the closed intervals: the first denominator is at least 13, its diagonal numerators are at least 6, the second denominator is at least 7, and the two numerator eigenvalues of L_F are at least 30. Directly,

$$\mathrm{Tr}\, L_E(s) = R_E(s), \qquad \mathrm{Tr}\, L_F(t) = R_F(t). \tag{26}$$

Now put

$$Q_E = I_2 \otimes L_E(s)^{\mathsf{T}} - \frac{A}{2}, \qquad Q_F = I_2 \otimes L_F(t)^{\mathsf{T}} - \frac{B}{2}. \tag{27}$$

Exact multiplication shows that both remainders annihilate $\mathrm{rvec}(I_2)$. On the complementary basis

$$\mathcal{B} = (e_{01}, e_{10}, e_{00} - e_{11}), \tag{28}$$

the two 3×3 Gram matrices have the six LDL^{T} pivots listed in Appendix A. All six are strictly positive throughout their respective closed intervals. Therefore $Q_E, Q_F \succeq 0$, each with kernel exactly $\mathrm{span}\{\mathrm{rvec}(I_2)\}$. Proposition 1 and (26) prove the radius identity, while (4) and (23) give (24).

Finally, $I_2 \otimes L_F^{\mathsf{T}}$ and $A/2$ are positive definite, so the kernels of the two summands in (14) are $\ker Q_E \otimes \mathbb{C}^4$ and $\mathbb{C}^4 \otimes \ker Q_F$. Their intersection is $\ker Q_E \otimes \ker Q_F$, spanned by $\mathrm{rvec}(I_2) \otimes \mathrm{rvec}(I_2)$. After applying P, the joint kernel is $\mathrm{span}\{\mathrm{rvec}(I_4)\}$. Unitarity fixes the proportionality constant to a phase, proving the uniqueness statement. □

5 Interpretation, scope, and limitations

The relative open subset

$$|s| < \frac{1}{5}, \qquad |t| < \frac{1}{5}, \qquad st \neq 0 \tag{29}$$

contains only pairs with two non-unital factors, as seen from (20). Theorem 3 therefore shows that two simultaneous output Bloch shifts do not, by themselves, produce an advantage for an entangling comparator in the inverse-Choi geometry. For this family the sixteen-dimensional joint problem reduces exactly to the two scalar expressions in (23).

The multiplier construction also has a methodological interpretation. Instead of optimizing directly over $U(4)$, one may seek local matrices whose traces equal the local maximally entangled radii and which dominate half of the corresponding inverse Choi matrices. Whenever they exist, the tensor remainder (14) is an explicit global certificate. This does not imply that such multipliers exist for arbitrary channels.

Several limits should be kept explicit. The result concerns one parallel use of two independent qubit channels. It is not about serial composition, continuous-time dynamics, or Gorini–Kossakowski–Sudarshan–Lindblad embeddability. All quantities are dimensionless and no continuum boundary conditions arise. The family was selected for exact tractability and is not claimed to represent generic non-unital noise. Most importantly, the universal multiplicativity conjecture for arbitrary full-rank qubit-channel pairs remains unresolved.

6 Exact reproducibility

The accompanying script `reproduce.py` reconstructs W, the two Choi families, their exact inverses, the magic-basis radii, the multipliers, all six interval certificates, the tensor-leg permutation, and the joint positive remainder using SymPy rational algebra. It also specializes to $s = 1/10, t = 1/8$, where

$$R_E(1/10)R_F(1/8) = \frac{573440}{5357}, \tag{30}$$

and verifies that the direct 16×16 remainder has rank 15. The generated JSON record contains exact expressions; any floating values are omitted from the proof path. Regeneration instructions and tested software versions are supplied in `REPRODUCIBILITY.md`.

Statements and declarations

Data and code availability. No empirical data are used. The exact symbolic checker, its generated JSON record, and regeneration instructions accompany this draft.

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A Closed-interval LDL^{T} certificate

Let

$$N(s) = 500s^3 + 75s^2 - 440s - 266. \tag{31}$$

An exact unpivoted LDL^{T} decomposition of $\mathcal{B}^{\mathsf{T}} Q_E \mathcal{B}$ has pivots

$$p_1^E = \frac{-N(s)}{(5s^2-6)(25s^2-14)}, \tag{32}$$

$$p_2^E = \frac{40(1250s^4 - 725s^2 + 294)}{(25s^2-14)N(s)}, \tag{33}$$

$$p_3^E = \frac{-100}{25s^2-14}. \tag{34}$$

For $|s| \le 1/5$,

$$5s^2 - 6 \le -\frac{29}{5}, \qquad 25s^2 - 14 \le -13, \tag{35}$$

$$N(s) \le 4 + 3 + 88 - 266 = -171, \tag{36}$$

$$1250s^4 - 725s^2 + 294 \ge 294 - 29 = 265. \tag{37}$$

These inequalities make all three pivots in (32)–(34) strictly positive.

Similarly, an exact decomposition of $\mathcal{B}^{\mathsf{T}} Q_F \mathcal{B}$ has pivots

$$p_1^F = \frac{10(25t^2+112)}{(25t^2-48)(25t^2-8)}, \tag{38}$$

$$p_2^F = \frac{-80(25t^2+32)}{(25t^2-8)(25t^2+112)}, \tag{39}$$

$$p_3^F = \frac{-20(625t^4 - 250t^2 + 96)}{(25t^2-8)(25t^2+32)}. \tag{40}$$

For $|t| \le 1/5$,

$$25t^2 - 48 \le -47, \qquad 25t^2 - 8 \le -7, \qquad 625t^4 - 250t^2 + 96 \ge 86. \tag{41}$$

All remaining displayed factors are positive, so the three pivots in (38)–(40) are also strictly positive. This completes the exact positivity step used in Theorem 3.

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